Homework 5. Push Down Automata. Turing Machine.

Submit PDF file (and Latex if you use it) of your solution to Canvas by 11:59 pm, Wed November 19. .

1. (15 - basics) (Push down automata). Given the PDA (push down automaton) for the palindrome (see lecture notes L5.2). Write the precise tuple form for this automaton. Remember to define each element of the tuple.

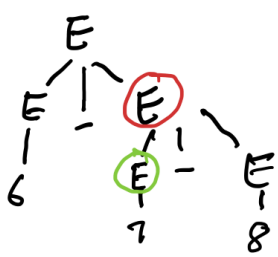
2. (10 - medium) (PDA) Design PDA for the language $L = \{0^n 1^n | n \geq 1\}$

3. (5 - advanced) (Comprehensive). Write a precise and general description of how a push down automaton can be constructed for a context free grammar G such that the automaton accepts same language represented/generated by G.

4. (12) (pumping lemma idea) Consider the CFG

   $E \rightarrow E - E$
   $E \rightarrow number$

   and the parse tree for $6 - 7 - 8$.



   We use the tree to show the intuitive ideas of pumping lemma.

   (a) Produce a new parse tree by replacing the subtree rooted at E with green circle by the subtree rooted at the E with red circle.

   (b) What is the yield of the new tree? Hint. Yield refers to the string/expression formed by the leaves of the tree in the order from the left to right.

   (c) Why the new tree is still a valid parse tree with respect to the CFG?

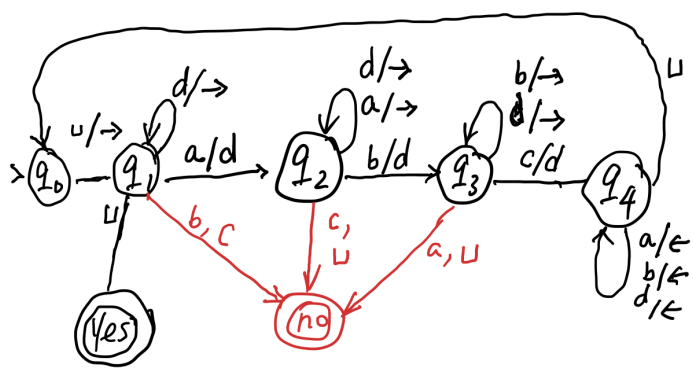
5. (8) (pumping lemma) Consider the language $L = \{a^i b^i c^i \mid i \geq 0\}$. Assume there exists CFG $G = (N, T, P, S)$ for L. Recall that $\phi(G)$ refers to the maximum number of terminals and non-terminals in the right side of any production rules.

   Let string $w = a^n b^n c^n$ where $n > \phi(G)^{|N|}$. By pumping lemma, there exists u, v, x, y, z such that $w = uvxyz$, either v or y is not an empty string, $|vxy| \leq n$, and for any k, $uv^k xy^k z \in L$.

   (a) Write a proof for the statement: if vxy does not contain c, then $uxz \in L$.

   (b) write a proof for $uxz \notin L$.

6. (28) (Turing Machine) Consider the Turing machine M in the picture below



   (a) What are the states of the machine?

   (b) What is the alphabet of the language that is accepted by this machine?

   (c) What are the halting states? (Halting states the nodes with double circle in the diagram.)

   (d) What is the initial state?

   (e) Write the *initial configuration* of M on string abc. (see definition of initial configuration in the slides).

   (f) Write the computation from $(q_1, \triangleright \sqcup \underline{d}ab)$ to $(q_3, \triangleright \sqcup dd\underline{d})$.

   (g) what is the value (as a tuple) of $\delta(q_2, b)$?

7. (5) (Church-Turing Thesis) What is Church-Turing thesis?

8. (10) (Turing Machine) Give a precise definition for "A Turing machine rejects a string" by assuming that the machine has only two halting states y and n. To make it easy for your writing to refer to the machine and string, you may give a name to each of them such as M and w respectively. Thus the concept becomes "A Turing machine M rejects a string w." You may find the definition of a similar concept of "accepts" in lecture note L6.1.

9. (Halting problem). What is halting problem? Give a proof that halting problem is not decidable. You are expected to apply the precise definition of halting problem, decideable and etc.